1.9.3 Properties

REVISED

- **Flexibility** is the paper's ability to withstand multiple folds before it breaks. This is important in printed products that may be folded and unfolded numerous times during use, such as books, maps and cartons.
- Paper made with virgin pulp, which has longer fibres, has a higher flexibility than recycled paper.
- **Printability** is how easy the paper can be printed on.
- The smoother and flatter the paper's finish, the easier it is to print on and the higher quality the print finish will be.
- **Biodegradability** means that the paper or board can break down and decompose without harming the environment.
- Normal paper and board will biodegrade quickly. Coatings on products such as coffee cups to make them waterproof significantly affect the length of time it takes the paper and board to decompose.

Flexibility: ability to bend and curve evenly and easily.

Printability: how easily a material can be printed on.

Typical mistake

Paper is not generally considered to be a strong material. However, for its weight it is a very strong material.

Now test yourself

TESTED

1 Name **one** use for tracing paper. (1)
2 Explain **two** reasons why corrugated cardboard is ideal for takeaway food packaging. (2)
3 Describe **one** use for mounting board. (2)

1.10 Thermoforming and thermosetting polymers

1.10.1 Thermoforming polymers

REVISED

- **Thermoforming polymers** soften when heated and can be moulded into shape.
- When the polymer cools it hardens and retains the shape.
- If the polymer is reheated it will try to return to its original shape.
- Thermoforming polymers can be reheated and reshaped numerous times.

Thermoforming polymer: a polymer that can be reheated and reshaped a number of times.

Acrylic

- Acrylic is a rigid, hard polymer usually in the form of a sheet.
- It is available in clear or a wide range of solid and translucent colours.
- Acrylic is easy to cut, bend, shape and polish.
- It has good **impact resistance** but is easily scratched.
- Common uses include as a substitute for glass, illuminated signs, aircraft canopies and car light clusters.

Typical mistake

Many students give acrylic as a polymer used for vacuum forming. Although it is easy to shape and bend using a strip heater, it is not suitable for vacuum forming.

Impact resistance: resistance to knocks and impacts.

High-impact polystyrene

- High-impact polystyrene (HIPS) is a thin, flexible and lightweight sheet polymer available in a wide range of colours.
- It is inexpensive and easy to cut, shape, bend and mould.
- HIPS is a popular choice for vacuum forming.
- Common uses for HIPS include disposable plates and cups, model kits, food containers and trays.

Biodegradable polymers

- Biodegradable polymers, or **biopolymers**, are produced from renewable raw materials such as corn starch, plant sugars and vegetable oil.
- They are environmentally friendly as they are compostable and biodegradable.
- Biopol® is a type of biopolymer. It is lightweight, hard and non-toxic.
- Common uses for biodegradable polymers include medical products such as dissolving thread for stitching wounds, bin bags, bottles and cups, and food packaging.

Biopolymer: a biodegradable polymer made from a plant source, such as corn starch or sugar beet.

1.10.2 Thermosetting polymers

REVISED

- **Thermosetting polymers** soften when heated and can be moulded into shape.
- During heating, a chemical change takes place.
- Once the thermosetting polymer has cooled, it hardens and sets in shape.
- Once set, the thermosetting polymer cannot be reheated and reshaped.
- If reheated, the thermosetting polymer will melt and burn.
- Thermosetting polymers cannot be recycled.

Thermosetting polymer: a polymer that can be heated and shaped only once.

Exam tip

Apart from biopolymers, polymers are not considered to be environmentally friendly materials and their use in modern society is causing considerable damage to the environment.

Polyester resin

- On its own, polyester resin is hard but very brittle and easy to snap.
- When combined and reinforced with fibreglass, it becomes GRP (glass reinforced plastic).
- GRP is very tough and has high impact resistance. It is also lightweight, waterproof and relatively easy to repair if damaged.
- Common uses for GRP include car body panels, sailing boats, canoes, corrugated roofing sheet, and electrical enclosures.

Urea formaldehyde

- Urea formaldehyde is a rigid, hard but brittle polymer.
- It is available in a wide range of colours.
- It has excellent heat resistance and is a good electrical **insulator**.
- Common uses for urea formaldehyde include electrical fittings (light switches, plug sockets), domestic appliance parts, laminate sheeting and wood glue.

Insulator: a material that does not conduct heat or electricity easily.

1.10.3 Properties

REVISED

The general properties of polymers are:

- **thermal insulator**: does not conduct heat easily
- **electrical insulator**: does not conduct electricity
- **toughness**: resistant to impact and wear.

Now test yourself

TESTED

1 Explain why thermoforming polymers are considered to be more environmentally friendly than thermosetting polymers. (2)
2 Give **two** reasons why polymers are often used for housing electrical equipment. (2)
3 Name a common use for HIPS. (1)

1.11 Natural, synthetic, blended and mixed fibres, and woven, non-woven and knitted textiles

- Fibres are very fine, hair-like structures that are spun (or twisted) together to make **yarn**.
- Yarns are woven or knitted together to create textile fabrics.

Yarn: spun threads used for knitting, sewing and weaving.

1.11.1 Natural fibres

REVISED

- Natural fibres come from natural sources.
- They are sustainable and biodegradable.

Table 1.10 Sources, properties and uses of natural fibres

Fibre	Source	Properties	Uses
Wool	Animal: sheep fleece	Warm, **absorbent**, low flammability, good **elasticity**, crease resistant	Warm outerwear (coats and jackets); knitwear; soft furnishings, carpets and blankets
Cotton	Vegetable: cotton plant/ seed pod	Absorbent, strong, cool to wear, hardwearing, creases easily, smooth, easy to care for, flammable, can shrink	Clothing, bedlinen, towels, sewing and knitting threads, soft furnishings

Absorbency: ability to soak up moisture.

Elasticity: ability to stretch and return to shape.

1.11.2 Synthetic fibres

REVISED

- Synthetic, or manufactured, fibres are artificial.
- Advantage: they can be manufactured to specific requirements.
- Disadvantage: they are not biodegradable and come from unsustainable sources.

Table 1.11 Sources, properties and uses of synthetic fibres

Fibre	Source	Properties	Uses
Polyester	Fossil fuels: coal/ petroleum	Strong when wet and dry, flame resistant, thermoplastic, hardwearing, poor absorbency	A versatile fibre used throughout textiles
Acrylic	Fossil fuels: petroleum	Strong except when wet, thermoplastic, burns slowly then melts, poor absorbency	Knitwear and some knitted fabrics, fake fur products including toys, upholstery and carpets

Exam tip

Look at the labels of different products. Try to identify the fibre used and work out what properties the fibre has that will allow the product to function as intended. The more you look at, the better your understanding of material selection will be.

1.11.3 Woven textiles

REVISED

- Woven fabrics consist of **warp** and **weft** yarns in an under/over configuration.
- Different **weave** creates fabrics with different **textures**, patterns and strength.
- **Plain weave** is the most basic weave. It is stable, strong and gives an even surface on both sides of the fabric. An example is calico.
- **Twill weave** gives a diagonal pattern and adds strength to the fabric. An example is denim.

Warp: yarn following the length of a fabric.

Weft: yarn running horizontally across a fabric.

Weave: the way yarns are laid in a pattern/formation in the construction of fabric.

Texture: the feel and finish of a material.

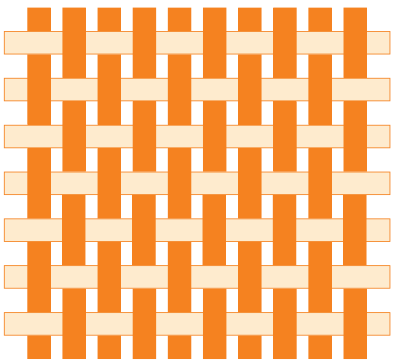


Figure 1.25 A plain weave structure

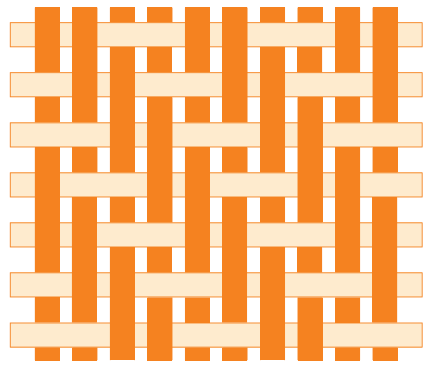


Figure 1.26 A twill weave structure

Typical mistake

Don't confuse warp and weft yarns used in fabric construction. Warp run the length of the fabric and weft run across the width of fabric from left to right.

1.11.4 Non-woven textiles

REVISED

Felted

- Felted fabrics are produced by applying moisture, heat, pressure and friction to a web of fibres, which causes the fibres to bind together. Wool and acrylic fibres are commonly used in this process.
- Felt is used in craft projects, accessories such as hats, and on the surface of snooker tables.

Bonded

- Non-woven or bonded textiles are **constructed** from a web of fibres, held together by **adhesive** or stitching.
- Most non-woven or bonded fabrics are not strong enough to be made into garments but are often used to reinforce woven and knitted fabrics.
- They are cheap to produce and are often used to make disposable products such as surgical gowns and masks, cleaning wipes and disposable protective suits.

Constructed: the way something has been made.

Adhesives: glues.

1.11.5 Knitted textiles

REVISED

- Knitted fabrics are made by creating a series of loops in the yarns that interlock together.
- There are two types of knitted fabrics: **warp knitting** (interlinking vertically) and **weft knitting** (interlinking horizontally).
- Knitted fabric is easy to stretch and warm to wear as the loops trap body heat.